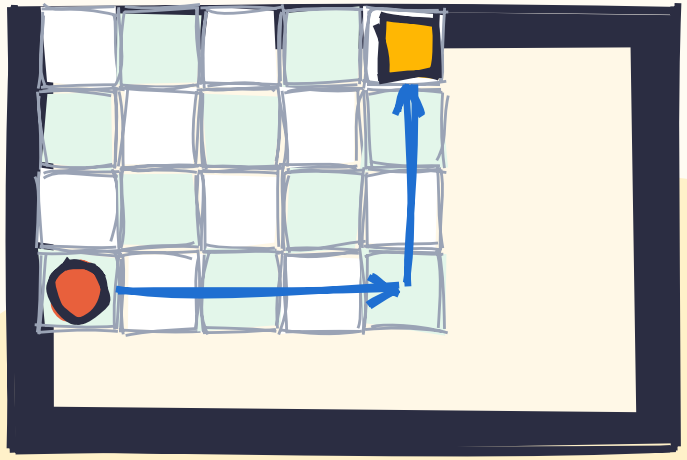
Little Explorers · Class 18

Code.org Puzzle Quest

Solve puzzles with sequences and loops

Sequences | Loops | Test & fix



Did you know?
Millions of children around the world learn to code with Code.org puzzles every year!

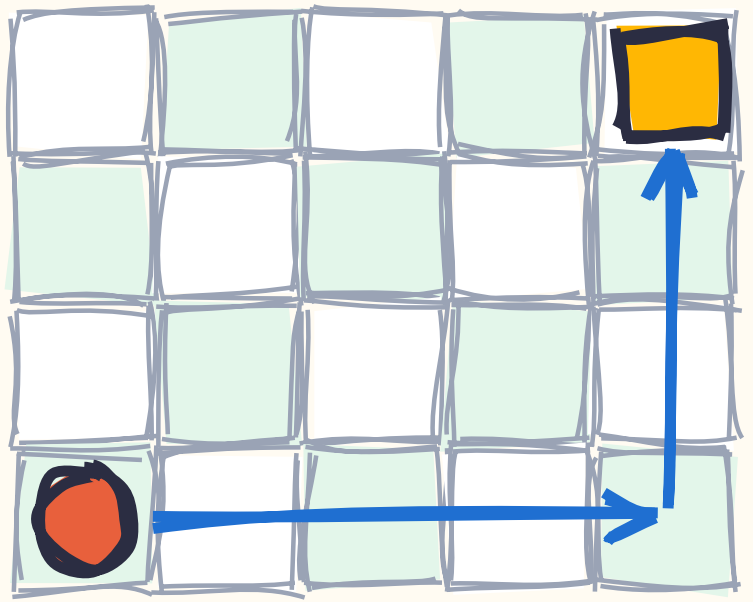
LEARN

Let's wonder...

How do we tell a computer where to go?

A lost bird

A little bird wants to reach the pig, but it can't think for itself. It only moves when **we** give it arrow blocks. How many blocks do we need? Which way?



Guess?

Shout at it?

Use arrow blocks!

Today: you solve **Code.org puzzles** with arrow blocks and discover a shortcut called a **loop**.

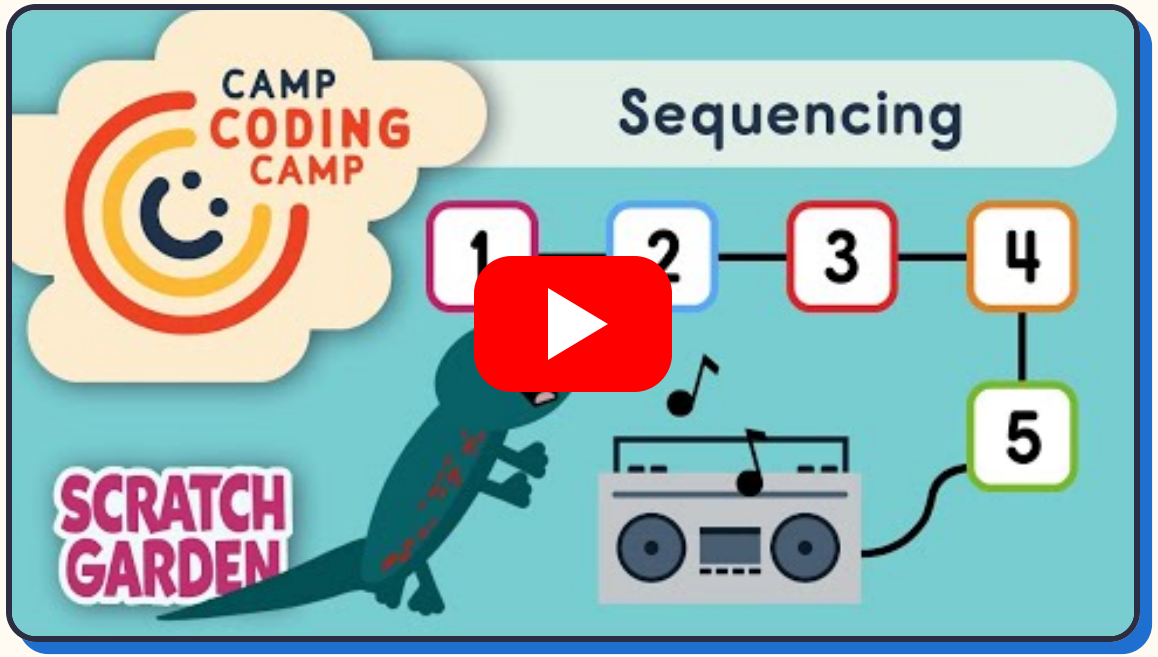
Learning objectives

Today I can...

1	**Sequence:** drag arrow blocks in a **sequence** to reach the goal	**Proof:** I finished 3 puzzles
2	**Loops:** use a **repeat** block instead of the same block many times	**Proof:** I used a repeat block in a puzzle
3	**Try again:** **test** my code and fix it when it goes wrong	**Proof:** I fixed a puzzle after it went wrong

Objective 1 · Sequence · video

Video time!



Sequencing

Scratch Garden · about 4 minutes · youtu.be/zW3YZdPmCnM

Watch for...

1. What does **sequence** mean?
2. What happens if the steps are mixed up?
3. Can you sing the sequence song steps?

Scan to watch

After the video: Stand up and do 3 steps in order: clap, jump, spin. Now do them backwards. Funny?

Objective 1 · Sequence

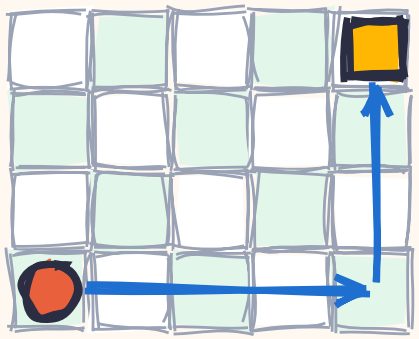
A sequence is blocks in order

The idea

Each block is one step. The computer does the **first** block, then the **next**, then the next.

Wrong order = wrong place! The bird might fly into a wall.

Example: hopscotch



In hopscotch you hop on 1, then 2, then 3. If you jump to 5 first, you've broken the game. Blocks work the same way: one, then the next.

LEARN

Objective 2 · Loops

Loops: repeat it!

The idea

If you need 'move forward' 5 times, you could drag 5 blocks… or use ONE **repeat 5 times** block!

A **loop** repeats the blocks inside it. Less dragging, fewer mistakes!

Example: a song chorus



In a song, the chorus comes back again and again. The songbook doesn't write it out every time: it just says 'repeat chorus'. That's a loop!

Real world

Loops everywhere



Clocks
Tick, tick, tick: repeat forever!



Songs
The chorus repeats



Washing machine
Spin round and round



Your heart
Beat, beat, beat, all day long

? Can you find a loop in your classroom?

Did you know?
Your heart beats about 100,000 times a day. That's a long loop!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Floor arrows
2. Start Code.org
3. Clap loops
4. Loop puzzles
5. Puzzle tracker
6. Puzzle challenges

Class game

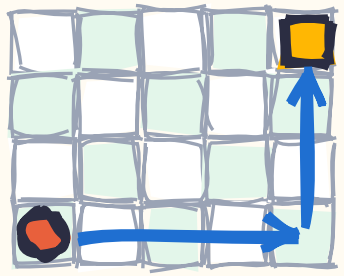
Floor arrows

How to play

1. Tape a 4×4 grid on the floor. Put a toy at the goal.
2. One child is the **bird**.
3. The team lays arrow cards in a row: ↑ → → ↑.
4. The bird follows the cards one by one. Did it reach the toy?

Then discuss

How many cards did you need? Did you use the same arrow many times?



Big idea

A sequence is a line of steps. The computer does them in order.

MAKE

Objective 1 · Build step 1 of 2

Start Code.org

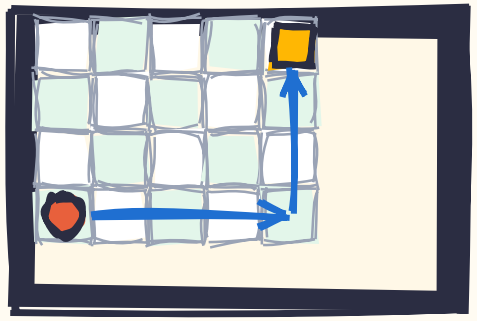
a Open **studio.code.org** and type the class code from the teacher.

b Click your name and your secret picture.

c Open **Course A** (or B). Start the **maze** puzzles.

d Drag arrow blocks under 'when run'.

e Press **Run**. Did it reach the goal?



Check: The bird reaches the pig!

MAKE

Class game

Clap loops

How to play

1. Teacher claps a pattern: clap, clap, stamp.
2. The class repeats it **3 times**.
3. Now a child says: 'repeat 4 times: jump, clap!'
4. Everyone does it together.

Then discuss

Was it easier to say 'repeat 4 times' or to say every step?



Big idea

A loop is a short way to say 'do it again'.

MAKE

Objective 2 · Build step 2 of 2

Loop puzzles

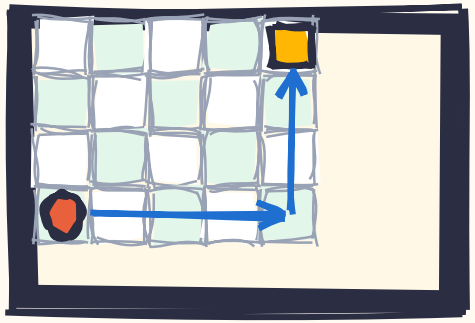
a Find the puzzles that show a **repeat** block.

b Put the move blocks **inside** the repeat block.

c Change the number: how many times?

d Press Run. Too far? Not far enough? Change the number!

e Try to finish with the **fewest** blocks.



Check: You solved a puzzle using a repeat block.

MAKE

Keep track!

Puzzle tracker

Puzzle	Blocks used	Tries
Puzzle 1		
Puzzle 2		
Puzzle 3		
Loop puzzle		

Talk about it

Which puzzle needed the most tries? What did you change?

Try, try again!
Programmers press Run many times.
Mistakes help you learn!

MAKE

Level up!

Puzzle challenges

Choose your challenge



Star puzzle: finish a puzzle with the fewest blocks.



Loop inside loop: can you put a repeat inside a repeat?



Buddy coding: one child drags, one child checks.



Next level: try the next lesson in your course.

Think like an engineer

Why is a loop better than 10 of the same block?

Boss level

Teach a younger child one Code.org puzzle!

Objective check

Can I really do it? Prove it.

01 Show a puzzle you solved with a sequence. ○ ○ ○

02 Show a puzzle with a repeat block. ○ ○ ○

03 Tell us one time your code went wrong and how you fixed it. ○ ○ ○

Rate yourself
Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

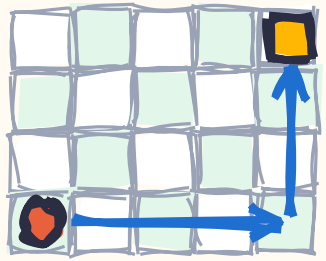
Teacher: Puzzle party: each child shows their best puzzle to a partner.

Show what you know

What did we learn?

- [x] A sequence is blocks in order
- [x] A loop repeats blocks
- [x] Press Run to test
- [x] Mistakes are OK: fix and try again

At home
Ask your family to do 'repeat 3 times: clap, jump'!



Exit ticket: what does a repeat block do?

Next class:
Program a robot car through a floor maze!